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SECTION B

Answer **all** *questions in the spaces provided.*

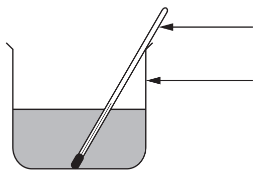
7. *(a)* A student was asked to find the enthalpy change of reaction, Δ*H*, for the thermal decomposition of calcium hydroxide.

$$Ca(OH)_2(s) \longrightarrow CaO(s) + H_2O(g)$$

This enthalpy cannot be measured directly so Hess's Law is generally used to calculate its value from the enthalpy changes for the reactions of calcium oxide and calcium hydroxide with hydrochloric acid.

The student added a known mass of calcium oxide to hydrochloric acid and measured the temperature change. This was repeated using a known mass of calcium hydroxide. In each experiment 50.0 cm^3 of 1.40 mol dm^{-3} hydrochloric acid was used.

The diagram shows the apparatus used.



When 1.90 g of calcium oxide was used a temperature rise of 20.5 °C was observed.

(i) The equation for the reaction of calcium oxide with hydrochloric acid is shown.

$$CaO(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + H_2O(l)$$

Use this equation to show that the hydrochloric acid was in excess in the reaction with calcium oxide. [3]

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(ii) Calculate the energy released in this reaction between calcium oxide and hydrochloric acid. [1]

Energy released = ... J

(iii) Calculate the molar enthalpy change for the reaction between calcium oxide and hydrochloric acid. [2]

$$CaO(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + H_2O(l)$$

Enthalpy change = ... $kJ\ mol^{-1}$
sign value

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(iv) The value of the molar enthalpy change of reaction for the reaction between calcium hydroxide and hydrochloric acid is –196 kJ mol^{-1}.

Use Hess's Law and your answer to part (iii) to calculate the enthalpy change of reaction for the decomposition reaction.

$$Ca(OH)_2(s) \longrightarrow CaO(s) + H_2O(g)$$

Show clearly how you obtained your answer.

(If you do not have an answer in part (iii), assume that the enthalpy change of reaction is –110 kJ mol^{-1}. This is **not** the correct value.) [2]

Enthalpy change = ... kJ mol^{-1}

(v) Suggest **two** changes to the apparatus shown that would give a more accurate value for the enthalpy changes of the reactions. Give a reason for your answer in both cases. [2]

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(b) Methanol, CH_3OH, is a liquid at room temperature and its molar enthalpy of combustion, $\Delta_c H$, can be found directly.

(i) Write the equation corresponding to the standard enthalpy change of combustion of methanol. [1]

....

(ii) Draw and label suitable apparatus as it is being used in an experiment to determine the enthalpy change of combustion of methanol. [2]

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(c) The equation for the combustion of ethene is shown.

$$C_2H_4(g) + 3O_2(g) \longrightarrow 2CO_2(g) + 2H_2O(g) \qquad \Delta_c H^\theta = -1387\ \text{kJ mol}^{-1}$$

(i) Use this and the values of the average bond enthalpies in the table to calculate the average bond enthalpy of C—H. [4]

Bond	Average bond enthalpy / kJ mol^{-1}
C═C	614
O═O	495
C═O	799
O—H	465

Average bond enthalpy = ... kJ mol^{-1}

(ii) State why the apparatus you have drawn in part *(b)* cannot be used to determine the enthalpy change of combustion of ethene. [1]

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